Experiment 7:. Random Sampling and Sampling Distribution

Objective: To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
# Population
population = np.arange(100, 201)
# Parameters
sample_size = 20
number_of_samples = 1000
# Store sample means
sample_means = []
# Random sampling
for i in range(number_of_samples):
    sample = np.random.choice(population, size=sample_size, replace=False)
    sample_means.append(np.mean(sample))
# Calculate mean of sample means
print("Mean of Sample Means:", np.mean(sample_means))
# Calculate standard deviation of sample means
print("Standard Deviation of Sample Means:", np.std(sample_means))
# Plot sampling distribution
plt.hist(sample_means, bins=30, edgecolor="black")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.title("Sampling Distribution of Sample Mean")
plt.grid(False)
plt.show()
```

OUTPUT:

```
Mean of Sample Means: 149.90145
Standard Deviation of Sample Means: 5.591058522095794
```

